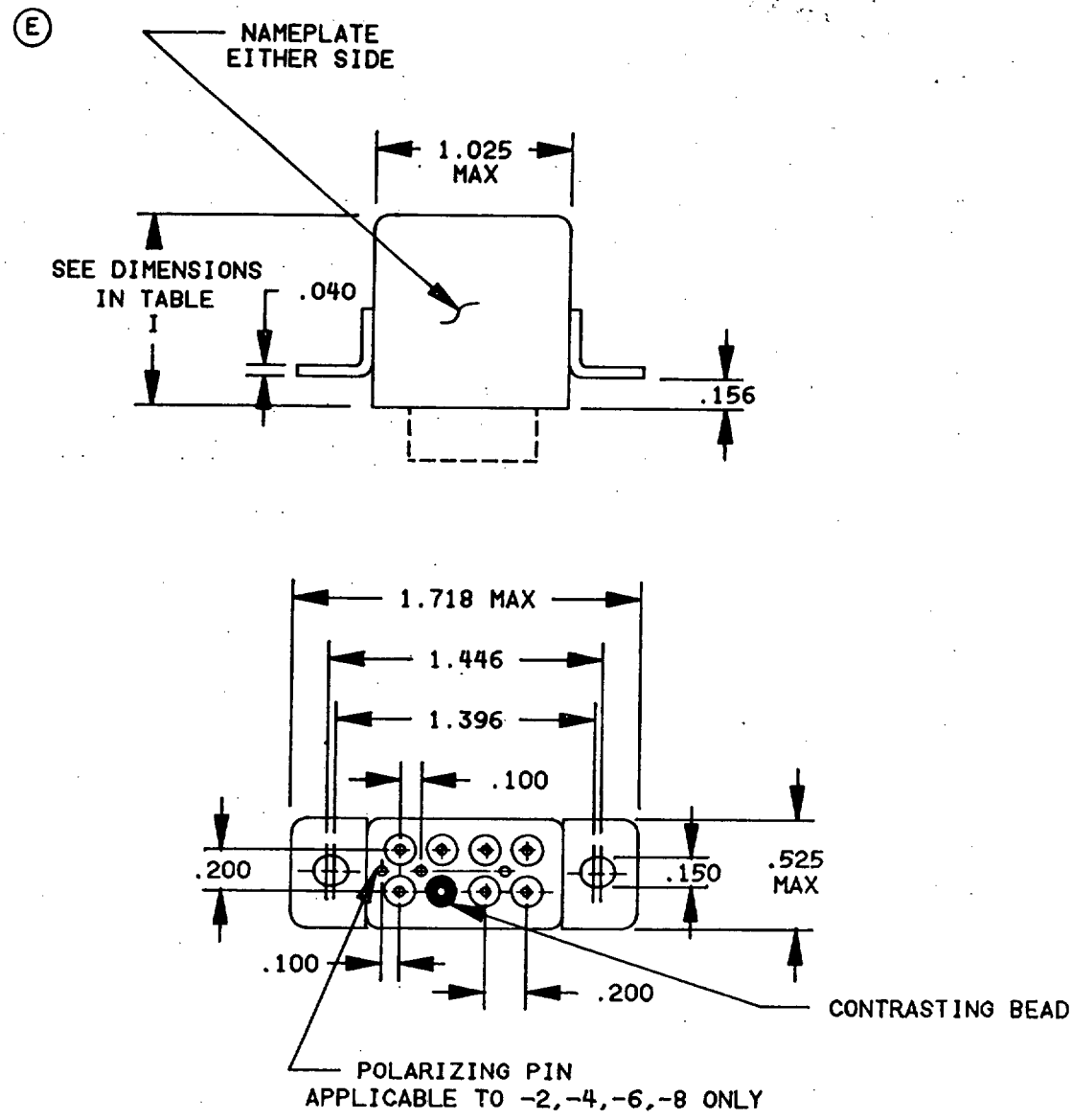
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USE MIL-R-6106/50

FED. SUP CLASS
5945



RAISED VERTICAL FLANGE MOUNT (DASH NUMBERS -1 THROUGH -8)

FIGURE 1. Outline drawing (for details see tables I and II).

(E) denotes changes

APPROVED 3 JUNE 71 REVISED (D) 28 MAY 85 (E) 29 APR 88

User activities: Army - Navy - Air Force -

Review activities: Army - Navy - EC Air Force - 99

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P.A AF-85 Other Cust NAVY-AS ARMY-ER	International interest	TITLE RELAYS, ELECTROMAGNETIC, MAGNETIC LATCH, 10 AMPERES, 2 PDT, ALL WELDED, HERMETICALLY SEALED	MILITARY STANDARD MS27744
Procurement Specification MIL-R-6106		SUPERSEDES:	PAGE 1 OF 7

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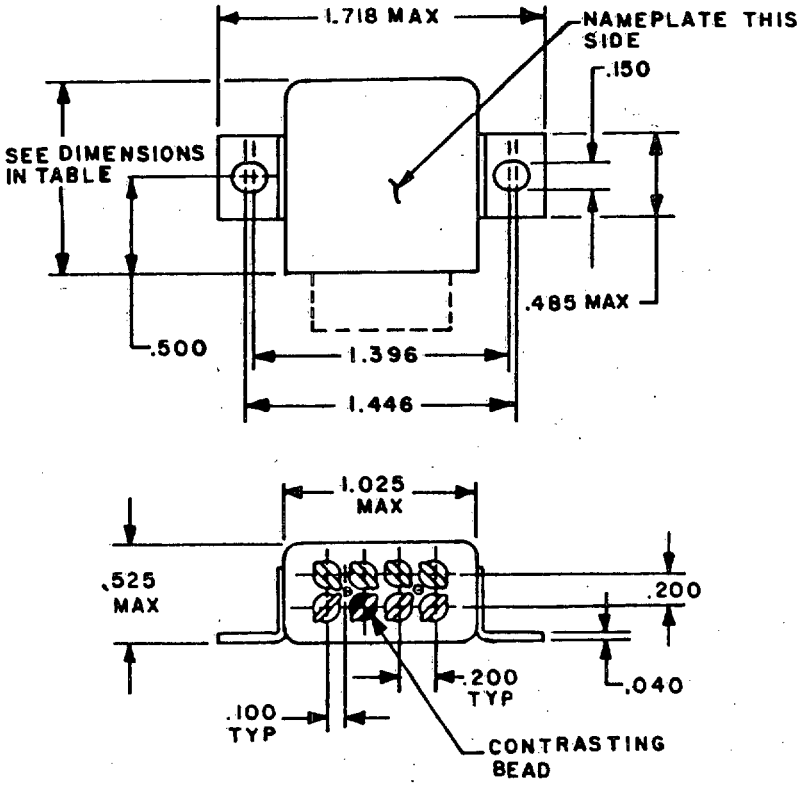
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5945-0762-1

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HORIZONTAL FLANGE MOUNT (DASH NUMBERS -9 AND -10)

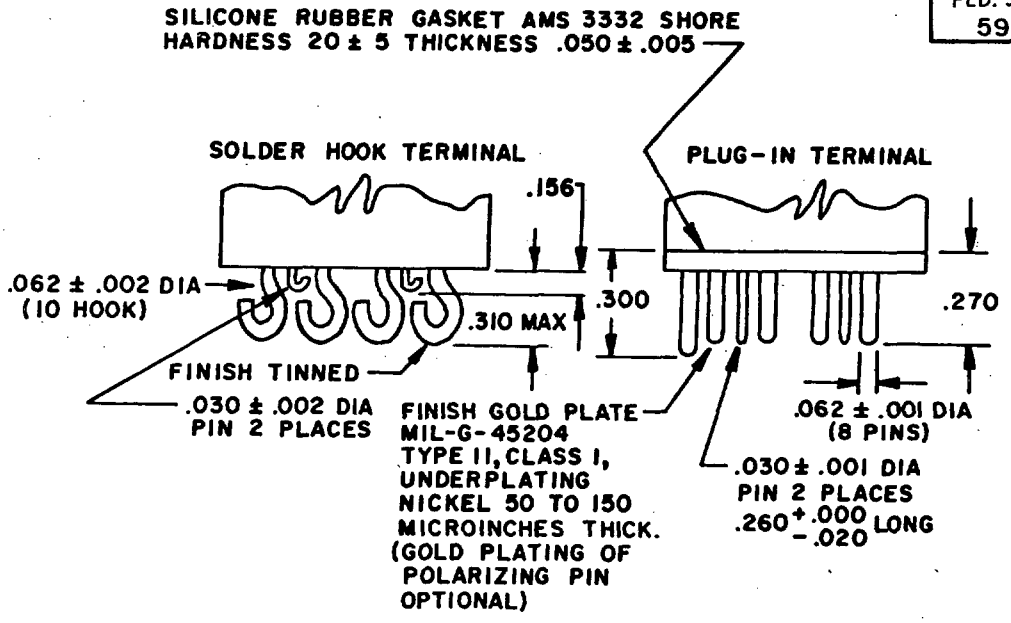
FIGURE 1. Outline drawing (for details see tables I and II) - Continued.

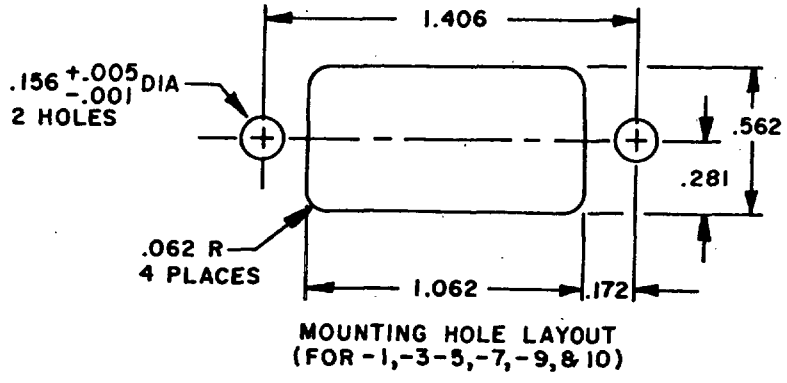
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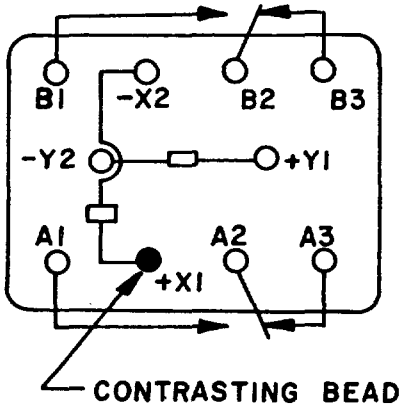
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MOUNTING HOLE LAYOUT
(FOR -1,-3-5,-7,-9,& 10)



CIRCUIT DIAGRAM TERMINAL VIEW

To close A1 and B1 contacts (operate) energize X1 and X2.
To close A3 and B3 contacts (reset) energize Y1 and Y2.

FIGURE 1. Outline drawing (for details see tables I and II) - Continued.

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INCHES	MM	INCHES	MM
.001	0.03	.260	6.60
.002	0.05	.270	6.86
.003	0.08	.281	7.14
.005	0.13	.300	7.62
.020	0.51	.310	7.87
.030	0.76	.485	12.32
.040	1.02	.500	12.70
.050	1.27	.525	13.34
.062	1.57	.562	14.27
.100	2.54	1.025	26.04
.150	3.81	1.062	26.97
.156	3.96	1.396	35.46
.172	4.37	1.406	35.71
.200	5.08	1.446	36.73
		1.718	43.64

NOTES:
1. Terminal numbers do not appear on the relay header, there shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
2. Dimensions are in inches.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm) for three place decimals.
4. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
5. Referenced Government documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this standard to the extent specified herein.
6. Relay is magnetically latched in both positions.
7. Caution note to observe polarity must appear on relays with dc coils.

FIGURE 1. Outline drawing (for details see tables I and II) - Continued.

TABLE I. Relay characteristics.

MS part number MS27744-	Type	Coil type	Terminal type	Mounting or mating socket	Dim max	Max weight in pounds
1	I	dc	Solder hook	N/A	1.010	0.09
2	I	dc	Plug-in	M12883/41-05,-10,-15,-20	1.010	0.09
3	I	ac	Solder hook	N/A	1.125	0.10
4	I	ac	Plug-in	M12883/41-05,-10,-15,-20	1.125	0.10
5	I	ac	Solder hook	N/A	1.010	0.09
6	I	dc	Plug-in	M12883/41-05,-10,-15,-20	1.010	0.09
7	I	dc	Solder hook	N/A	1.125	0.10
8	I	ac	Plug-in	M12883/41-05,-10,-15,-20	1.125	0.10
9	I	dc	Solder hook	N/A	1.010	0.09
10	I	ac	Solder hook	N/A	1.125	0.10

1/ and 1/ } 1/

(E) 1/ High shock and vibration.

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APPROVED 3 JUNE 71

User activities: Army - Navy - Air Force -
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TABLE II. Operating characteristics.

MS part no. MS27744	Coil data										Time-milliseconds-max				
	Coil	Nominal			Max		Max pick-up voltage 1/			Drop out voltage 1/	Operate	Release	Bounce		
		Volts	Freq Hz	Res Ω	Volts	Amp	Normal 1/	High temp test	Cont current test				Main NO	Main NC	Aux NO	Aux NC
-1,-2,-5,-6,-9	X1, X2	28	dc	-	29	.05	18	19.8	22.5	-	10		1	1	-	-
	Y1, Y2	28	dc	-	29	.05	18	19.8	22.5	-	10		1	1	-	-
-3,-4,-7,-8,-10,	X1, X2	115	400	-	122	.04	90	95.4	103.5	-	20		1	1	-	-
	Y1, Y2	115	400	-	122	.04	90	95.4	103.5	-	20		1	1	-	-

1/ Over temperature range.

TABLE III. Rated contact load (amperes per pole) case grounded.

Type of load	Life operating cycles x 10^3	28 V dc				115 V ac, 1 phase				115/200 V ac, 3 phase				See appropriate notes
		Main		Aux		Main		Aux		Main		Aux		
		NO	NC	NO	NC	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	400 Hz	60 Hz	
Resistive	100	10	10	-	-	10		-	-	10		-	-	1/
Inductive	100													
Inductive	20	8	8	-	-	8		-	-	8		-	-	
Motor	100	4	4	-	-	4		-	-	4		-	-	
Lamp	100	2	2	-	-	2		-	-	2		-	-	
Transfer, load														2/
Mechanical life reduced current	400	2.5	2.5	-	-	2.5		-	-	2.5		-	-	
Min current	Applicable per specification													

1/ Resistive rating 12 amperes for MS part no. -5, -6, -7, and -8.
2/ Transfer load indicates relay suitable for transfer between unsynchronized ac power supplied at rating indicated.

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ENVIRONMENTAL CHARACTERISTICS	MS part no. -1, -2, -3, -4, -9, -10	MS part no. -5, -6, -7, -8
Temperature range	-70 C to +125 C	
Max altitude rating	300,000 ft	
Shock G-level	200 g's	500 g's
Duration	6 ms	1 ms
Max duration contact opening	10 µs	
Vibration-sinusoidal		
G-level	30 g's	50 g's
Frequency range	10 through 3,000 Hz	
Vibration-random		
Applicable spec	MIL-STD-810	
Power spectral density	0.4 g^2/Hz	0.65 g^2/Hz
Frequency range	50 through 2,000 Hz	
Acceleration	15 g's	

(E) ELECTRICAL CHARACTERISTICS

	MS part no.	
Insulation resistance, initial		
After life or environmental tests		
Dielectric strength (sea level)		
Coil to coil and coil to case	Initial: 1000 V rms	After life test: 1000 V rms
Aux contacts	Initial: —	After life test: —
All other points	Initial: 1250 V rms	After life test: 1000 V rms

Dielectric strength (altitude)1/	80,000 ft	300,000 ft
Coil to coil and coil to case	250 V rms	500 V rms
Aux contacts	—	—
All other points	350 V rms	500 V rms

Max contact drop initial		0.150 volt
After life test		0.175 volt
Overload current	40 amperes dc	60 amperes ac
Rupture current	50 amperes dc	80 amperes ac
Duty rating		Continuous

RFI specification
(Applicable to coil circuits of ac operated relays)

(E) Quality conformance inspection:
Performance of groups B and C inspection are not applicable.

(E) Supersession data: See table IV.

1/ Dielectric may be improved by suitable insulation of terminals and wiring after installation.

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Ⓔ Table IV. Supersession data. 1/

Superseded part number MS27744 -	Superseding part number M6106/50-
1	001
2	002
3	007
4	008
5	No superseding dash number
6	"
7	"
8	"
9	003
10	009

1/ This table will also be used for logistic support.

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Procurement Specification MIL-R-6106		SUPERSEDES:	PAGE 7 OF 7

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